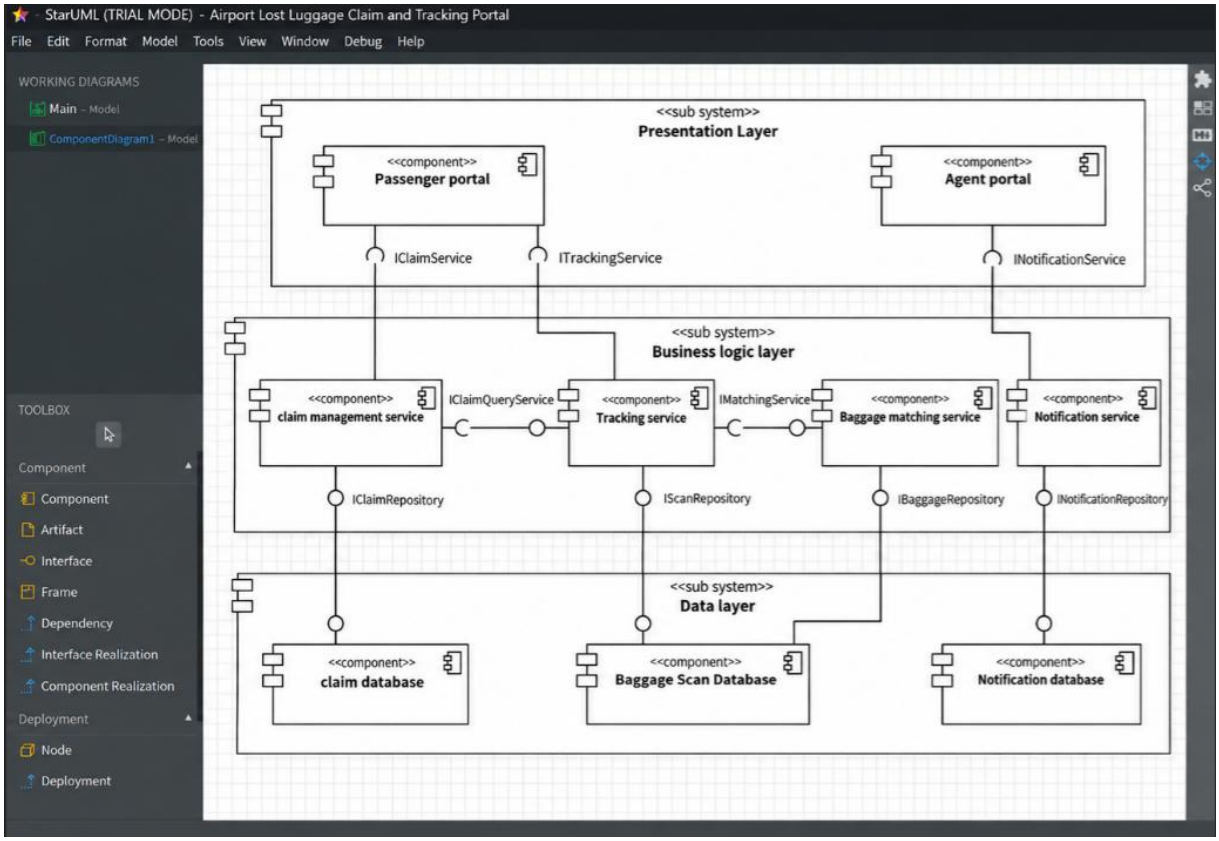
Lab 3: Component Modelling And Architectural Pattern Selection

TOPIC :AIRPORT LOST LUGGAGE CLAIM AND TRACKING PORTAL

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Component Diagram Explanation

The component diagram shows the main parts of our Airport Lost Luggage Claim and Tracking Portal and how they communicate with each other. The system is divided into three layers: the Presentation Layer, Business Logic Layer, and Data Layer, ensuring efficient communication between users, services, and databases.

1. Presentation Layer

The Presentation Layer provides the interfaces through which passengers and airport staff interact with the system.

Passenger Portal:

The Passenger Portal allows passengers to submit lost luggage claims and track the status of their baggage. It communicates with the Claim Service for claim-related operations and the Tracking Service to retrieve baggage updates.

Agent Portal:

The Agent Portal is used by airport staff to manage claims and assist passengers. It also communicates with the Notification Service to send important updates regarding baggage status and claim progress.

2. Business Logic Layer

The Business Logic Layer contains the core services responsible for processing claims, tracking baggage, matching luggage, and sending notifications.

Claim Management Service:

This service handles the creation, updating, and management of lost luggage claims. It stores claim information in the Claim Database through the Claim Repository.

Tracking Service:

The Tracking Service retrieves baggage scan information from the Baggage Scan Database and provides passengers with the latest tracking status. It also communicates with the Claim Management Service through the Claim Query Service.

Baggage Matching Service:

This service compares baggage scan records with reported claims to identify potential matches between lost luggage and scanned baggage. It accesses baggage information through the Baggage Repository.

Notification Service:

The Notification Service sends alerts and updates to passengers and airport staff whenever there is a change in baggage status or claim progress. It stores notification records in the Notification Database through the Notification Repository.

3. Data Layer

The Data Layer stores all the information required by the application.

Claim Database:

It stores passenger claim details, claim status, and other claim-related information.

Baggage Scan Database:

It stores baggage scan records collected from airport checkpoints, allowing the Tracking and Baggage Matching Services to retrieve scan information.

Notification Database:

It stores notification history and delivery records for messages sent to passengers and airport staff.

System Workflow

The system works as follows:

Passenger → Submit Lost Luggage Claim → Claim Management Service → Claim Database → Tracking Service → Baggage Scan Database → Baggage Matching Service → Notification Service → Passenger and Agent Receive Status Updates.